Many Daughters: Pre-analysis Plan

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The pre-analysis plan below details the exact way we will test each of the four hypotheses.

1 Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

1.1 Construction of Gender Equality Index

The SOEP data does not include a direct measure of gender support. Following Wandschneider et al. (2022), we construct a gender index ranging from 0 to 1 as the dependent variable. Wandschneider and colleagues screened SOEP questionnaires comprising more than 3,200 variables to identify gender-related indicators. They excluded variables that: (i) directly measured health or gender inequalities; (ii) did not describe gender processes, practices, or attitudes; (iii) showed no significant association with sex assigned at birth; (iv) had more than 25% missing cases. This selection process resulted in 29 variables, as listed in Wandschneider et al. (2022) Supplementary Material 7. However, eight variables from that list (noted as *bip** variables in SOEP) were not available to us. We, therefore, used a subset of 21 variables as the starting point for our analysis to calculate gender equality index. We further dropped five variables with more than 50% of missing values (in the 5% sample) and three variables that were highly correlated with other variables.

Table 1 presents the final set of 13 questions used for this project which we further divide into two subgroups — seven variables referring to affective relations (emotional resources), used to predict gender equality as measured by attitudes (to obtain an index for H1) and six variables referring to economic and power relations (access to resources and participation), used to predict gender equality as measured by behaviors or outcomes (to construct an index for H3 later in this document).

Table 1: **Questions**

Question	**SOEP code**
Affective relations (emotional resources)	
Worried about finances	$plh0033$
Worried about consequences from climate change	$plh0037$
Worried about crime in Germany	$plh0040$
Satisfaction with housework	$plh0174$
Satisfaction with personal income	$plh0176$
Satisfaction with amount of leisure time	$plh0178$
Willingness to take risks	$plh0204_h$
Economic and power relations (access to resources and participation)	
Working experience part-time employment	$pgexpft$
Paid work in last 7 days	$plb0018$
Hours weekdays housework	$pli0043_h$
Hours weekdays care for persons	$pli0046$
Hours weekdays repairs	$pli0049_h$
Hours weekdays leisure, hobbies	$pli0051$
Variables dropped	
Worried about environment	$plh0036$ (high correlation with $plh0037$)
Worried about peace	$plh0038$ (high correlation with $plh0037$ and $plh0040$)
Worried about global terrorism	$plh0039$ (more than 50% missing values)
Worried about own retirement pension	$plh0335$ (more than 50% missing values)
Satisfaction with standard of living	$plh0162$ (more than 50% missing values)
Nowadays can't trust anyone	$plh0193$ (more than 50% missing values)
Most people are exploitative vs. fair	$plh0195$ (more than 50% missing values)
Working experience full-time employment	$pgexppt$ (high correlation with $pgexpft$)

Notes: Variables used to construct Gender equality index. Adapted from Wandschneider et al. (2022). Variables categorized as referring to affective relations (emotional resources) are used to construct gender equality index as measured by attitudes — H1. Variables categorized as referring to economic and power relations (access to resources and participation) are used to construct gender equality index as measured by behavior and outcomes — H3. The last panel — Variables not used are variables categorized to one of the above mentioned groups by Wandschneider et al. (2022) but not used in the construction of our index due to reason specified.

For H1, we use the first seven questions to estimate a hierarchical logistic regression model using *.melogit* command, with sex at birth as the dependent variable. The model includes seven variables related to affective relations as predictors, and a random intercept individual lel (pid). Some variables contain missing values, and we aim to retain observations even when some regressors are missing. To address this, we assign a value of zero to missing variables, allowing these observations to be included in the regression. Additionally, we create so-called shadow variables for each regressor: these take a value of 1 if the original variable is missing and 0 otherwise. Shadow variables are included in the

model together with regressors to capture the effect of missingness. This approach ensures that observations with partial data remain in the analysis, enabling the non-missing variables to contribute to the results.

The gender index is estimated as the conditional probability to be categorized as "female" ranging from 0 to 1. A higher value on the index indicates stronger alignment with gendered attitudes typically associated with women. We code this index as $genderscore2$.

1.2 Inclusion/Exclusion Criteria

We apply several inclusion and exclusion criteria to define the appropriate testing sample:

1. Drop parents with no information on the birth year of a child.
2. Drop parents with children born before the start of the survey period + one year, i.e. children born before the second survey year.
3. Drop parents with children when the boy is born before the girl.
4. Keep parents in the sample until the year their first son is born.

We do not apply any additional restrictions to the sample. Note that we use the same exclusion/inclusion criteria throughout the project, i.e. for H1-H4.

1.3 Hypothesis testing

In settings with multiple units that adopt treatment at different times — known as staggered treatment adoption — the standard Difference-in-Differences (DID) approach is often implemented using a two-way fixed effects (TWFE) regression. The workhorse specification is[1]

$$y_{it} = \alpha + \lambda_t + \beta_{FE} D + e_{it}. \tag{1}$$

Here, the dependent variable y_{it} is the gender equality index measured by attitudes from 0 to 1, (variable $genderscore2$), for each person (parent) i in survey time t. The

[1]The notations are simplified to capture a more general static form.

variable of interest is the treatment measure, D, expressed as a year the first daughter is born (variable $firstgirl$). The specification includes a full set of year dummies — λ_t.

For the main estimation, we rely on the Callaway and Sant'Anna (2021) staggered diff-diff estimator. We use STATA18 command *.csdid* with option *notyet* that allows us to include alongside with 'never treated' group, a 'not-yet-treated' group as control group. The command provides/estimates the ATT for (i) all groups across all periods; (ii) each group or cohort, across all periods; (iii) each period, across all groups or cohorts; and, (iv) the dynamic ATT's — estimated for each period relative to the period first treated, across all cohorts. For our analysis we use the dynamic ATT, which estimates the effect pre- and post-treatment. To test the hypothesis, we estimate the post-treatment effect. The positive and significant coefficient confirms the hypothesis. The positive and significant estimate based on the 5% subsample confirms the hypothesis that a parent exhibits more pronounced feminine social practices after the birth of a daughter.

The model is estimated on unbalanced panel. By default, we use robust standard errors.

2 Hypothesis 2: Having daughters leads to more left-leaning political preferences

2.1 Dependent variable — Political Left-Right Scale

We use a question that captures respondents' self-placement on the political spectrum, ranging from left (liberal/progressive) to right (conservative): "In politics, people often talk about 'left' and 'right'. Where would you place your views on this scale, where 0 means left and 10 means right?" As this question is not asked in every survey year, we assume relative stability in political orientation and impute missing values using responses from the most recent previous wave.

2.2 Inclusion/Exclusion Criteria

We use the same inclusion/exclusion criteria as for the H1 test. Please, see details in section 1.2 .

2.3 Hypothesis testing

Similarly to the H1 test, we estimate the model, eq. (1), with Callaway and Sant'Anna (2021) estimation technique. The dependent variable y_{it} is the political left-right scale — $leftright2$ — ranging from 0 for the left and 1 for the right. The post-treatment effect is the ATT estimate of interest. The negative and significant coefficient beta confirms the hypothesis. On average, the person will have more left-leaning political views after the first daughter is born.

3 Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)

3.1 Dependent variable — Gender equality index

Following Wandschneider et al. (2022), we construct a gender index ranging from 0 to 1 as the dependent variable. See section 1.1 above.

For H3, we use the set of questions from the second panel in Table 1 and estimate a hierarchical logistic regression model with sex at birth as the dependent variable. The calculated conditional probability to be categorized as "female" ranging from 0 to 1 that we interpreted as the gender equality index. A higher value on the index indicates stronger alignment with gendered practices typically associated with women. We code this index as $genderscore1$. Each parent is assigned an index. To have an index at the household level, we take the mean score per households — *hid*.

3.2 Inclusion/Exclusion Criteria

We use the same inclusion/exclusion criteria as for the H1 test. Please, see details in section 1.2. Since the household identifier does not uniquely identify set of parents, we keep only observations with one set of parents per household. Over the years, the construct of the household might change, resulting in changing records for the year of birth of first girl. We use the year of the 'older' daughter of the household as a treatment period.

3.3 Hypothesis testing

Similarly to the H1 test, we estimate the model, eq. (1), with Callaway and Sant'Anna (2021) estimation technique. The dependent variable y_{it} is the gender equality index measured by behavior and outcomes — $genderscore1$ — ranging from 0 to 1, 1 being more female oriented for each household i at survey period t. Note that i now is reserved for household not a parent. The post-treatment effect is the ATT estimate of interest. The positive and significant coefficient beta confirms the hypothesis. On average, the household will have more gender-equal behavior and outcomes.

4 Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)

4.1 Dependent variable — Gender equality index

Following Wandschneider et al. (2022), we construct a gender index ranging from 0 to 1 as the dependent variable. See section 1.1 above. The index is calculated for each parent.

For H4, we use the same dependent variable a for H1 — gender equality measured by attitudes. Recall that a higher value on the index indicates stronger alignment with gendered attitudes typically associated with women and is coded as $genderscore2$.

4.2 Inclusion/Exclusion Criteria

We use the same inclusion/exclusion criteria as for the H1 test. Please, see details in section 1.2.

4.3 Hypothesis testing

Similarly to the H1 test, we estimate the model, eq. (1), with Callaway and Sant'Anna (2021) estimation technique. The dependent variable y_{it} is the gender equality index measured by attitudes — *genderscore*2 — ranging from 0 to 1, 1 being more female oriented. The post-treatment effect is the ATT estimate of interest.

The twist here is that we estimate the same model separately for a subsample of male parents and female parents. To test for difference in attitudes, we implement a block-bootstraping technique. We do 100 iterations on randomly selected 1000 parents, collect the estimates. Based on the sample of 100 estimates for females and 100 estimates of males, we calculate the mean difference in coefficients, calculate standard errors and p-values.

Let $\hat{\beta}_{m,i}$ and $\hat{\beta}_{f,i}$ denote the bootstrapped estimates of the coefficient from the i-th iteration for the male and female subsamples, respectively, and N be the number of bootstrap iterations — 100.

- Mean Difference:

$$\bar{d} = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{\beta}_{m,i} - \hat{\beta}_{f,i} \right)$$

- Standard Deviation of the Differences:

$$s_d = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} \left(d_i - \bar{d} \right)^2} \quad \text{where } d_i = \hat{\beta}_{m,i} - \hat{\beta}_{f,i}$$

- Standard Error of the Mean Difference:

$$\text{SE}_{\bar{d}} = \frac{s_d}{\sqrt{N}}$$

- t-statistic:

$$t = \frac{\bar{d}}{\mathrm{SE}_{\bar{d}}}$$

- Two-sided p-value:

$$p = 2 \cdot P(T_{N-1} > |t|)$$

References

Callaway, B. and P. H. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

Wandschneider, L., O. Sauzet, O. Razum, and C. Miani (2022). Development of a gender score in a representative german population sample and its association with diverse social positions. *Frontiers in Epidemiology 2*, 914819.